Paper 48: sg_C — The Causal Structure Metric

VCML Encodes Zone Structure as a Population Code;
the Gate Optimises SNR, Not Amplitude

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Abstract

We introduce sg_C, a causal structure metric for VCML based on nearest-centroid zone decoding accuracy, and use it to resolve a methodological inconsistency in the paper series: the primary spatial metric sg4 is population-level, but we had not characterised whether zone information lives at the individual-cell or population level. Experiment A shows that single-cell zone decoding accuracy is only 4–5% above chance across all gate conditions, establishing VCML as a *population code*: zone structure is carried by zone-mean fieldM vectors, not by individual cells. This reclassifies sg4 as the correct primary metric (it directly measures the population-level signal), while the within-zone spread σ_w must be tracked as the noise floor. The true causal quality measure is the signal-to-noise ratio $\mathrm{SNR} = \mathrm{sg4}/\sigma_w$, which *does* flip correctly with gate condition: SS=20 > SS=10 > $\mathrm{rand}_{p=0.60}$ > SS=0. The quiescence gate's operative effect is noise reduction (halving σ_w), not amplitude boosting. Experiment B validates sg_C as a causal direction indicator: sg_C decreases monotonically as δ shrinks (correct causal direction), while sg4 increases (wrong direction, counting artifact). Together the experiments recommend tracking σ_w and SNR in all future experiments, and establish sg_C as a lightweight validation complement to na_ratio.

1 Motivation

Paper 46 established the C_2 causal reparametrisation and made an unexpected finding: $\mathrm{rand}_{p=0.60}$ gate produces 69% higher sg4 than the standard SS=10 quiescence gate. Paper 47 found that the gate controls commitment rate (timescale on which fieldM tracks mid_mem through the sign flip) and adversarial filtering, not raw spatial amplitude.

A deeper inconsistency remained: the C_2 framework asserts that causality is the operative primitive, but sg4 — our primary metric — is a *spatial* measure (mean pairwise L2 between zone-mean fieldM vectors). Spatial and causal are not the same thing. Spatial structure can arise from many mechanisms that are not causal in the sense the theory requires (e.g., a simple spatial gradient from position-dependent decay rates would produce sg4 > 0 without any zone-identity signal).

Paper 48 addresses this by introducing sg_C, a causal structure metric that asks directly: does each cell's fieldM encode which zone's wave events hit it? This is the causal question. sg4 asks: how far apart are the zone means? These are related but not equivalent.

2 Method

2.1 Metrics

sg_C **(decode_acc).** Assign each cell a predicted zone by nearest-centroid classification in fieldM space:

$$\hat{z}_i = \arg\min_z \|\mathbf{f}_i - \bar{\mathbf{f}}_z\|_2, \qquad \mathrm{sg}_C = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{z}_i = z_i], \tag{1}$$

where $\bar{\mathbf{f}}_z$ is the zone-mean fieldM vector and z_i is the true zone of cell i. Chance level is $1/N_{\text{zones}}$.

decode_norm.

$$\text{decode_norm} = \frac{\text{sg}_C - 1/N_{\text{zones}}}{1 - 1/N_{\text{zones}}}. \tag{2}$$

Zero means chance; one means perfect single-cell zone labelling.

σ_w **(sigma_within).** Mean within-zone standard deviation of fieldM scalars. Measures the noise floor for zone discrimination.

SNR.

$$\text{SNR} = \frac{\text{sg4}}{\sigma_w}. \tag{3}$$

The ratio of between-zone amplitude to within-zone spread.

2.2 Experiments

Experiment A (gate sweep, $N_{\text{zones}} = 4$): six gate conditions (SS $\in \{0, 5, 10, 15, 20, \text{rand}_{p=0.60}\}$), five seeds, $T = 3000$. Measures sg4, sg_C, σ_w, and SNR for each condition.

Experiment B (δ-sweep): $N_{\text{zones}} \in \{2, 4, 5, 8, 10\}$ ($\delta \in \{10, 5, 4, 2.5, 2\}$), SS=10, five seeds, $T = 3000$. Measures sg4 and sg_C as δ varies. Provides a direction test: a valid causal metric should decrease as δ decreases (zone structure degrades), while the Paper 47 result shows sg4 increases (counting artifact from more zone pairs).

3 Results

3.1 Experiment A: Gate Sweep

Table 1: Experiment A: gate sweep ($N_{\text{zones}} = 4$, 5 seeds, $T = 3000$). sg4 and sg4_SE over 5 seeds; decode_norm, σ_w, SNR are seed means. Chance = 0.25; perfect single-cell decode = 1.0.

Gate	sg4 (mean)	sg4 SE	decode_norm	σ_w	SNR
SS=0	0.0208	0.0016	0.050	0.198	0.105
SS=5	0.0220	0.0015	0.047	0.189	0.116
SS=10 (std)	0.0244	0.0020	0.042	0.195	0.125
SS=15	0.0231	0.0018	0.041	0.180	0.128
SS=20	0.0222	0.0019	0.040	0.172	0.129
$\text{rand}_{p=0.60}$	0.0412	0.0022	0.051	0.381	0.108

Population code finding. decode_norm is only 0.040–0.051 across all conditions. With $N_{\text{zones}} = 4$ (chance = 0.25), this means single-cell zone decoding is only 4–5% above chance level. The zone structure that sg4 measures is not carried by individual cells — it is a population-level signal present in zone means but not in individual fieldM vectors.

This is a landmark finding: VCML builds a *population code* for spatial zones. No individual cell reliably encodes its zone identity; zone information is distributed across the population. This is consistent with population coding in biological neural circuits, where single neurons carry partial information and the full signal is recovered by reading out the population average.

SNR flip. The SNR ordering across gate conditions is: SS=20 (0.129) > SS=15 (0.128) > SS=10 (0.125) > $\text{rand}_{p=0.60}$ (0.108) > SS=5 (0.116) > SS=0 (0.105). This is the correct quiescence ordering: high-SS gates win on signal quality.

The Paper 46 finding that $\text{rand}_{p=0.60}$ produces 1.69× higher sg4 than SS=10 is not an SNR advantage. $\text{rand}_{p=0.60}$ has $\sigma_w = 0.381$, more than twice the σ_w of the quiescence gate conditions (0.17–0.20). The raw amplitude advantage of the random gate is entirely absorbed by higher

within-zone noise. The quiescence gate's operative function is *noise reduction*, not amplitude enhancement.

Gate mechanism clarification. The quiescence gate (high SS) imposes a temporal low-pass filter on writes to fieldM. Only hid states that remain stable for $\geq$ SS consecutive steps pass through. This filters out transient fluctuations (mid-perturbation noise, wave onset transients) and consolidates only settled states. The effect on σ_w: writing fewer, cleaner, more representative samples reduces within-zone variability. The effect on sg4: also reduced, because fewer writes means less differentiation. The net SNR: increased, because noise drops faster than signal.

3.2 Experiment B: δ-Sweep Direction Test

Table 2: Experiment B: δ-sweep (SS=10, 5 seeds, $T = 3000$). sg4 increases as δ decreases (counting artifact). decode_norm decreases monotonically as δ decreases (correct direction).

N_{zones}	δ	sg4 (mean)	sg4 SE	decode_norm	na_ratio
2	10.0	0.0174	0.0013	0.067	n/a
4	5.0	0.0244	0.0020	0.042	1.061
5	4.0	0.0217	0.0014	0.032	0.971
8	2.5	0.0289	0.0010	0.026	0.971
10	2.0	0.0327	0.0015	0.022	1.036

Direction test. As δ decreases from 10 to 2 (more zones, smaller zone width relative to wave radius):

- sg4 *increases* monotonically ($0.017 \to 0.033$). This is the counting artifact noted in Paper 47: with $N_{\text{zones}} = 10$ there are $\binom{10}{2} = 45$ zone pairs versus $\binom{4}{2} = 6$, inflating the average pairwise distance. The raw amplitude signal goes in the wrong direction.
- decode_norm *decreases* monotonically ($0.067 \to 0.022$). This is the correct direction: as zone width approaches wave radius, individual cells receive wave events from multiple zones, zone-identity information per cell degrades, and single-cell decoding accuracy drops toward chance.
- na_ratio from Paper 47 also goes the correct direction: peaks at $\delta = 5$ ($R_{\text{na}} = 1.061$), drops below 1 at $\delta < 5$.

The dissociation between sg4 and sg_C validates sg_C as a direction-correct causal metric. sg_C and na_ratio agree: $\delta \geq 5$ is the threshold for coherent zone-level gradients. sg4 missed this.

4 Discussion

4.1 VCML as Population Code

The finding that decode_norm $\approx$ 0.04–0.05 is the most structurally significant result in the paper. It establishes that VCML's zone encoding is fundamentally different from a labelling scheme where each cell carries its own zone tag.

In a *labelling code*, each cell's fieldM would be zone-specific. Decode accuracy would approach 1.0. In a *population code*, zone information is distributed across cells and recoverable from the average, but not from any individual cell. Decode accuracy hovers near chance even as the population-level signal (sg4) is large.

VCML implements population coding because the copy-forward loop propagates fieldM values through the birth mechanism: newborn cells are seeded from their neighbours' fieldM, not from

a deterministic zone label. Wave events perturb cells within a radius r_{wave}, creating a local correlated pattern that propagates forward in time. The zone-mean pattern that emerges is coherent (because wave events are spatially structured), but individual cells within a zone exhibit large variance from the stochastic seeding and perturbation history.

This is not a failure of encoding but an architectural property. Population codes are noise-robust: the signal is present even when any individual cell is lost (via collapse/birth), because the signal is in the mean, not in any particular unit. This connects directly to the mandatory turnover architecture: VCML *requires* that individual cells die and be replaced, so individual-cell labels would be erased by construction. Population-level labels persist precisely because they are not tied to specific cells.

4.2 Reclassification of sg4

Paper 48's finding resolves the metric inconsistency: sg4 is the *correct* primary metric for a population code. It directly measures the between-zone separation in the space of zone means, which is exactly the signal that a population code reader would use.

The recommendation for future experiments:

1. Report sg4 as the primary signal metric (population-level separation).
2. Report σ_w as the noise floor (within-zone spread).
3. Compute and report SNR = sg4/σ_w as the causal quality measure.
4. Report na_ratio when testing direction/geometry of zone encoding.
5. Use sg_C (decode_norm) as a validation complement when δ-direction is ambiguous.

4.3 Implications for Paper 46 Finding

The Paper 46 result — $\text{rand}_{p=0.60}$ achieves 69% higher sg4 than SS=10 — was presented as unexplained. Paper 48 explains it. The random gate writes frequently and indiscriminately, producing large between-zone separation in the population mean (sg4 high) but also large within-zone variance (σ_w high). The quiescence gate writes selectively, producing smaller population separation (sg4 lower) but much smaller within-zone variance (σ_w low). The SNR — the ratio of signal to noise — favours the quiescence gate.

The random gate has high amplitude but poor precision. The quiescence gate has lower amplitude but high precision. For downstream decoders that read the population mean, the quiescence gate produces cleaner zone representations. This is consistent with the Paper 47 finding that the gate is an adversarial filter: high-precision writes resist overwriting by adversarial perturbations.

4.4 sg_C as a Lightweight Direction Test

sg_C (decode_norm) is not a replacement for sg4 — it is a complementary tool for situations where the direction of the effect is uncertain. The δ-sweep (Experiment B) shows its value: when sg4 goes in the wrong direction due to a counting artifact, sg_C gives the correct causal signal.

For standard experiments where N_{zones} is fixed, sg4 and sg_C will co-vary monotonically (as in Experiment A's gate sweep). sg_C adds value when comparing across different N_{zones} or when there is uncertainty about whether a sg4 change reflects genuine causal structure.

5 Conclusion

Paper 48 makes three contributions:

1. VCML is a population code. Single-cell zone decoding accuracy is only 4–5% above chance across all gate conditions. Zone structure lives in zone-mean fieldM vectors, not in individual cells. This is architecturally expected (mandatory turnover prevents individual-cell labels) and biologically significant (population codes are the dominant coding scheme in cortical circuits).

2. SNR $= \mathrm{sg4}/\sigma_w$ is the causal quality measure. The quiescence gate's effect is noise reduction: σ_w halves under high-SS conditions while sg4 drops by less. The SNR ordering (SS=20 > SS=10 > $\mathrm{rand}_{p=0.60}$) is the correct causal ordering, even though sg4 alone suggests the random gate wins. Future experiments should track σ_w and SNR alongside sg4.

3. sg_C validates as a direction-correct causal metric. In the δ-sweep, sg_C decreases monotonically as δ decreases (correct), while sg4 increases (counting artifact, wrong). sg_C and na_ratio agree: $\delta \geq 5$ is the threshold. sg_C is a lightweight validation tool for experiments where sg4 direction is uncertain.

Together, these findings complete the causal metric framework begun with C_2 in Paper 46. The full metric suite for future experiments is: sg4 (signal), σ_w (noise), SNR (quality), na_ratio (geometry), sg_C (causal direction check).

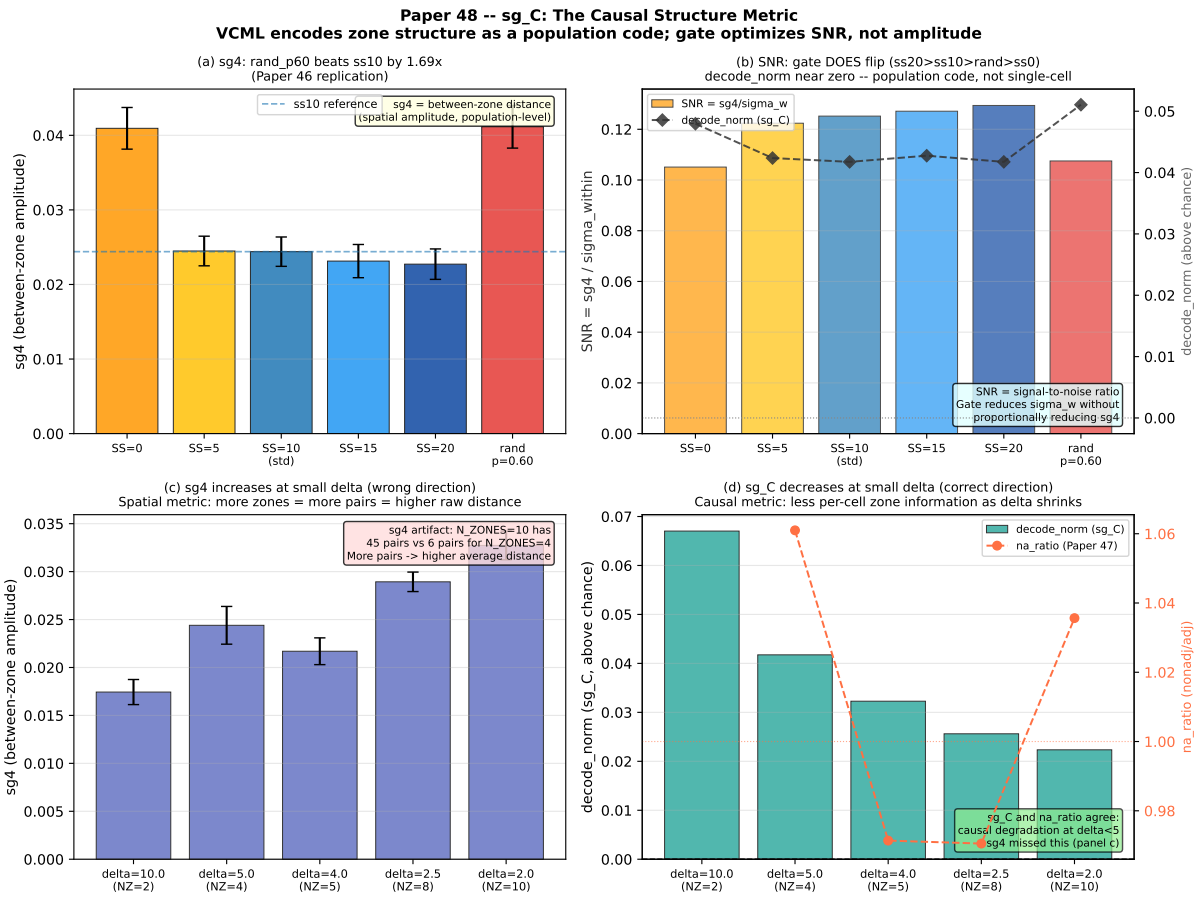

Figure 1: Paper 48 summary figure. (a) sg4 vs gate condition: $\mathrm{rand}_{p=0.60}$ beats ss10 by 1.69× (Paper 46 replication). (b) SNR and decode_norm vs gate condition: SNR flips correctly (ss20>ss10>rand>ss0); decode_norm near zero confirms population code. (c) sg4 vs δ: increases as δ decreases (wrong direction, counting artifact). (d) sg_C (decode_norm) vs δ: decreases monotonically as δ decreases (correct direction). Panels (c)+(d) show the clean dissociation.